Byungjun Kim

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About Me

Hello, I am Byungjun Kim, an AI Research Engineer at Smilegate, specializing in AI-driven NPC systems. I hold a Master's degree in Artificial Intelligence and a Bachelor's degree in Computer Engineering from Chung-Ang University, under the supervision of Professor Bugeun Kim.
My research focuses on analyzing behavioral patterns exhibited by large language models and understanding the underlying mechanisms that drive these behaviors. Through this work, I aim to enhance the performance and adaptability of LLM-based agents in specialized domains, particularly in interactive and environment-aware systems.

Research Fields

- LLM-based Applications
- Behavioral Patterns of LLMs in specific environments, including social deduction games, and role-playing games
- Human-like utterance generation using psychological basis

Education

MS **Chung-Ang University**, Artificial Intelligence — Mar 2024 – Dec 2025
- **Coursework:** Natural Language Processing, Machine Learning, AI

BS **Chung-Ang University**, Software Engineering — Mar 2019 – Aug 2023
- Summa Cum Laude
- **Coursework:** Computer Science, Algorithm, Network

Work History

AI Research Engineer, Smilegate — South Korea, Dec 2025 - Present
- Advance AI-driven mercenary system in Lost Ark Mobile
- Explore and prototype methodologies for applying language models to NPC systems and the virtual worlds composed of such NPCs

Experience

University of Birmingham, Visiting Researcher — United Kingdom, Sep 2025 - Nov 2025
- Studyed neural text decoding using LLM, supervised by Prof. Hyojin Park

Hyundai NGV Boost-Campus, Mentor — Korea, Jul 2025 - Sep 2025
- Confidential project

Department of AI, Chung-Ang University, Teaching Assistant — Korea
- Mobile programming (Department of AI at Chung-Ang University)
- Heuristic Algorithm (Department of AI at Chung-Ang University)

ELU Lab, Chung-Ang University, Research Intern — Korea, May 2023 – May 2024
- Researched financial QA
- Investigated behavioral patterns of LLMs in abnormal environments, including obscured communication.

NAVER Connect Boostcamp AI, Camper — Korea

- Completed an external training program in the NLP domain. During the program, I participated in internal competitions and worked on an industry-collaborative project. Sep 2022 – May 2024

On-campus AI club CUAI, Member Korea

- Participated in NLP study sessions and a summer project as a member of CUAI. Mar 2022 – Aug 2022

Publications

Fine-Grained and Thematic Evaluation of LLMs in Social Deduction Game Sep 2025
IEEE Access

Byungjun Kim, Dayeon Seo, Minju Kim, Bugeun Kim

Leveraging Large Language Models for Active Merchant Non-player Characters Aug 2025
Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence AI, Arts & Creativity.

Byungjun Kim*, Minju Kim*, Bugeun Kim

DART: An AIGT Detector using AMR of Rephrased Text Apr 2025
Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies

Hyeonchu Park*, ***Byungjun Kim****, Bugeun Kim

PHISH in MESH: Korean Adversarial Phonetic Substitution and Phonetic-Semantic Feature Integration Defense (arXiv. preprint) May 2025

Byungjun Kim, Minju Kim, Hyeonchu Park, Bugeun Kim

Projects

LLM-based Agents Simulating Psychological Phenomena in Human Communication thingsflow and ELU Lab

- Explore how psychological mechanisms behind human utterance generation can be applied to LLMs to enhance persona-consistent language generation.
- Domain: LLM, Psycology
- Role: Project manager (ELU Lab)

Transformer-based Vision Model Compression Nota AI and NAVER Connect Boostcamp AI

- Modifying the architecture of the SegFormer-b2 to achieve model compression and computational efficiency.
- Domain: Model compression
- Role: Camper researcher (Boostcamp AI)

Skills

Language

- Korean - Native
- English - TEPS (387, 2+), TOEIC (875)

Technologies: computer science, python, pytorch

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